VIT

Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

Reg. Number:	

Continuous Assessment Test (CAT) – II SEPTEMBER 2026

Programme	:	B. Tech. in Electronics Engineering (VLSI Design and Technology)	Semester	:	Fall 26-27
Course Code & Course Title	:	BEVD302L Principles of Communication Systems	Class Number	:	CH2026270101957
Faculty	:	Ralph Thangaraj	Slot	:	F1+TF1
Duration	:	90 Minutes	Max. Mark		50

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information
- If any data point seems to be missing in the question, make an assumption on the value, justify it and solve.
- Only non-programmable calculator without storage is permitted

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	BT Level
1	a	A message signal is given by $m(t) = 3\cos(2000\pi t) + 5\cos(6000\pi t) + \sin(8000\pi t)$ (i) Determine the Nyquist rate for $m(t)$. **[2 Marks]** (ii) If $m(t)$ is sampled at $f_s = 7$ kHz, determine whether aliasing occurs. **[1 Mark]** (iii) If aliasing occurs, determine the aliasing (folded) frequency at which the 4 kHz component will appear in the sampled signal. **[2 Marks]**	5	3	2
	b	A borehole water-level sensor sends readings to a data-logging hub 2 km away over a dedicated shielded twisted-pair cable. The cable route runs alongside heavy industrial machinery for part of its length resulting in electromagnetic interference. The system shares a reference clock which is distributed to all the sensors, so reliable timing is available. The sensor node runs with limited battery back-up, so minimizing the transmitter's average power consumption is a priority. Which one of the three schemes - Pulse Amplitude Modulation (PAM), Pulse Width Modulation (PWM) or Pulse Position Modulation (PPM), will you choose, as the best suited scheme, for this link? Support your answer with a complete, structured analysis. i. Analyse the scenario and identify every technical requirement from the scenario. **[6 Marks]** ii. Systematically reject the two unsuitable schemes, giving a	15	4	3

		technical reason for rejecting each. [**6 Marks**] iii. Justify as to why the remaining scheme is the best match, and briefly note any remaining limitation of your chosen scheme. [**3 Marks**]			
2	a	Explain the concept of a Correlation Receiver in digital demodulation. How does the concept of Signal Space (basis functions) simplify the design and analysis of this receiver?	5	5	2
	b	A satellite downlink is designed to transmit high data rate signals. The on-board transmit amplifier is power-limited (not bandwidth-limited) and its response becomes more nonlinear as the input signal's peak amplitude rises, so large envelope variations in the transmitted signal are undesirable. The link operates at low to moderate SNR. You may NOT use binary ASK, binary FSK, binary PSK or QPSK. Design a signal constellation that meets the above requirements. i. Analyse the scenario and list out the design requirement it implies for the constellation. [**6 Marks**] ii. Consider at least two candidate constellation geometries for $M \geq 8$, reject the one that violates a requirement identified above, with clear reasoning for rejecting that. [**6 Marks**] iii. Present your final constellation as a diagram with neatly labeled coordinates for a stated average symbol energy of E_S. Calculate its minimum Euclidean distance, and justify as to why it is the best available match to the scenario. [**3 Marks**]	15	5	3
3		A source generates 10 symbols with the following probabilities: $P = [0.35, 0.15, 0.12, 0.10, 0.08, 0.07, 0.06, 0.04, 0.02, 0.01]$ (i) Construct a Shannon–Fano code for this source. Show the binary tree with each step and the list the resulting code neatly in a table. [**6 Marks**] (ii) Calculate the average codeword length L, and the source entropy H. [**4 Marks**]	10	6	4

**********All the best **********